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COLOUR COATING OF COMPRITOL® 888 ATO MATRIX TABLETS: A FEASIBILITY STUDY

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GATTEFOSSÉ

INTRODUCTION

Compritol® 888 ATO is a lipid excipient with long saturated fatty acid chains (C22). It has a high melting point (70°C) and a low hydrophilic-lipophilic balance value (HLB 1). Therefore Compritol® 888 is used as lubricant and matrix former for sustained release tablets[1]. Given the lipophilic nature of the surface of these tablets questions have been raised concerning the feasibility of coating such tablets with hydrophilic polymers. The purpose of this study was to color coat Compritol® 888 matrix tablets and to evaluate the adherence and appearance of the polymer film on the tablet surface.

EXPERIMENTAL METHODS

Tablet compression

40% Compritol® 888 ATO (glyceryl behenate NF), 36% Emcompress Premium (DCP), 23.5% Supertab® SD 11 (lactose) and 0.5% magnesium stearate were blended and compressed using a single station press with 9 mm biconvex tooling (60 tablets/min).

Tablet coating

750g placebo tablets were loaded in a pan coater (Loedige HCT 30 mini) and sprayed with a 15% (w/w) aqueous solution of Opadry® red (HPMC 2910, FD&C Red #40, TiO_2, PEG 400) to 5% coating level by weight. Samples were taken at 1, 2, 3 and 4% coating weight gain. The coating conditions are listed in Table 1. The coated tablets were further dried in the pan for 10 min at 60°C inlet temperature.

Tablet characterization

The weight uniformity of uncoated and coated tablets was determined for each batch using 20 tablets. Tablets were tested for friability and hardness using standard methods.

Table 1: Coating conditions.

Parameters	Nozzle diameter	Drum speed	Atomization pressure	Mean spray rate	Inlet temperature	Product temperature
	0.8 mm	14 rpm	1.5 bar	2.6 g/min	60 °C	29 °C

RESULTS AND DISCUSSION

Placebo tablets of the investigated formula had a mean weight of 287.8 ± 3.28 mg, relatively low hardness (19.9 ± 2.8 N) and high friability (0.39%). Nevertheless, tablets did not break or cap during the coating process. The total coating time was 1h40.

Throughout the coating process the tablet weight increased very evenly (Figure 1a) and the practical weight gain achieved was almost that the theoretical (Figure 1b).

Figure 2 shows photographic images of the uncoated and coated placebo tablets. The aqueous polymer coating solution properly adhered to the placebo cores, demonstrated by the formation of a continuous and homogeneous film on the tablet surfaces and the central band. The film coat was smooth and showed no sticking defects irrespective of the coating level. The minimum required coating weight gain was 3%. This is within the general requirement of 3 to 4% to achieve adequate opacity of the film coat. It is noteworthy that the hardness of 19.9 ± 2.8 N for uncoated tablets increased to 48.4 ± 3.0 N for tablets coated with 3% weight gain, and to 66.8 ± 4.9 N for tablets coated with 5% weight gain.

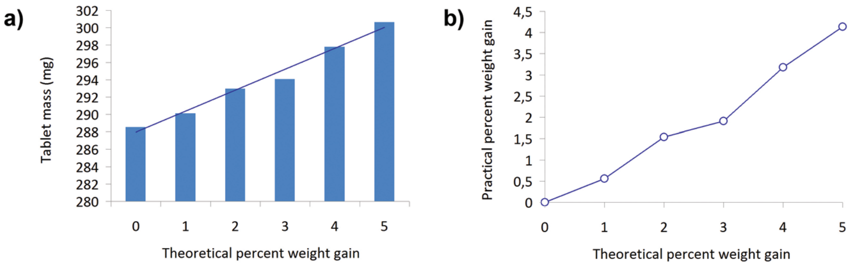

Figure 1: Evolution of: a) the tablet weight and b) the percent coating weight gain.

Figure 2: Photographic images of uncoated and coated Compritol® 888 placebo tablets. The coating weight gain is indicated in the figure.

CONCLUSION

The aqueous film coating of lipophilic Compritol® 888 ATO tablets with a hydrophilic coating polymer Opadry® was straightforward. The coating solution adhered correctly to the tablet core and resulted in smooth, defect free coating. Tablets were not damaged during the coating process despite the relatively low tablet strength. The lipophilic character of these tablets does hence not prevent polymer adhesion to the tablet surface.

REFERENCES

[1] Roberts, M. et al. Drug Dev Ind Pharm. 2012, 38, 1068-76.